$AR(1): y_t = a_0 + a_1 y_{t-1} + \varepsilon_t \qquad \varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$

$AR(p): y_t = a_0 + a_1 y_{t-1} + \dots + a_p y_{t-p} + \varepsilon_t$

$MA(1): y_t = a_0 + \varepsilon_t + \beta_1 \varepsilon_{t-1}$

$MA(q): y_t = a_0 + \varepsilon_t + \beta_1 \varepsilon_{t-1} + \dots + \beta_q \varepsilon_{t-q}$

$$y_t^{ogo} = C_1 \lambda_1^t + C_2 \lambda_2^t = C_3 + \lambda_2^t + C_4 \lambda_3^t + C_5 \lambda_3^t t + \dots$$

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$$y_t - \mu_y = \sum_{i=0}^{\infty} \beta_i \varepsilon_{t-i} \qquad \varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$$

$$\sum_{i=0}^{\infty} |\beta_i| < \infty$$

$$\sum_{i=0}^{\infty} a_i y_{t-i} = a_0 + \varepsilon_t \leftarrow \text{обратимость процесса.}$$

$$\sum_{i=0} |a_i| < \infty$$

$$\left[1 + aL + a^2 L^2 + \dots\right] y_t = \left[\frac{1}{1-aL}\right] y_t$$

$$\rho_s = \frac{\gamma_s := Cov(y_t, y_{t-s})}{\sqrt{Var\, y_t \; Var\, y_{t-s}}} = \frac{\gamma_s}{\sqrt{\gamma_0^2}} = \frac{\gamma_s}{\gamma_0}$$

($Var\, y_t$ и $Var\, y_{t-s}$ под корнем равны γ_0)

$$y_t = 1 + 1{,}2\, y_{t-1} - 0{,}35\, y_{t-2} + \varepsilon_t$$

$$y_t - 1{,}2\, y_{t-1} + 0{,}35\, y_{t-2} = 1 + \varepsilon_t$$

$$\left[1 - 1{,}2L + 0{,}35L^2\right] y_t = 1 + \varepsilon_t$$

$$\lambda^2 - 1{,}2\lambda + 0{,}35 = 0$$

$$\lambda = 0{,}7 \qquad (\lambda - 0{,}7)(\lambda - 0{,}5)$$
$$\lambda = 0{,}5$$

$$1 - 1{,}2z + 0{,}35z^2 = 0$$

$$\Delta = 1{,}44 - 4 \cdot 0{,}35 = 1{,}44 - 1{,}4 = 0{,}04$$

$$z_1 = \frac{1{,}2 + 0{,}2}{0{,}7} = \frac{2}{=}$$

$$z_2 = \frac{1{,}2 - 0{,}2}{0{,}7} = \frac{1}{0{,}7} = \frac{10}{7}$$
$$=$$

$$y_t^* = 1{,}2\, y_{t-1}^* - 0{,}35\, y_{t-2}^* + \varepsilon_t$$

$$E(y_t^* y_{t-1}^*) = E\left[\left(1{,}2 y_{t-1}^* - 0{,}35 y_{t-2}^* + \varepsilon_t\right) y_{t-1}^*\right] =$$

$$\gamma_1 =$$

$$= 1{,}2\gamma_0 - 0{,}35\gamma_1 + \underbrace{E(\varepsilon_t y_{t-1}^*)}_{=0}$$

$$\gamma_1 = 1{,}2\gamma_0 - 0{,}35\gamma_1$$

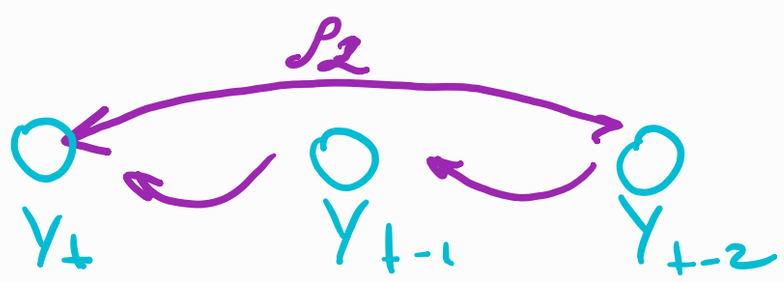
$$\boxed{\gamma_1 = \frac{1{,}2}{1{,}35}\gamma_0} \qquad \rho_1 = 1{,}2/1{,}35$$

$$\gamma_2 = E(y_t^* y_{t-2}^*) = 1{,}2\gamma_1 - 0{,}35\gamma_0$$

$$\rho_2 = 1{,}2\frac{\gamma_1}{\gamma_0} - 0{,}35$$
$$\rho_1$$

$$\gamma_s = 1{,}2\gamma_{s-1} - 0{,}35\gamma_{s-2} \Rightarrow \rho_s = 1{,}2\rho_{s-1} - 0{,}35\rho_{s-2}$$

$$\rho_s \xrightarrow[s \to \infty]{} 0$$



φ_{ss}: $y_t = \varphi_{0s} + \varphi_{1s} y_{t-1} + \varphi_{2s} y_{t-2} + \dots + \underline{\varphi_{ss}} y_{t-s} + v_t$

$\rho_\tau = \varphi_{1s}\rho_{\tau-1} + \varphi_{2s}\rho_{\tau-2} + \dots + \varphi_{ss}\rho_{\tau-s}, \quad \tau = \overline{1,s}$

φ_{11}: $\varphi_{11} = \rho_1$

$y_t = \varphi_{01} + \varphi_{11} y_{t-1} + v_t$

$\rho_1 = \varphi_{11}\rho_0 \Rightarrow \rho_1 = \varphi_{11}$ ($\rho_0 = 1$)

φ_{22} : $y_t = \varphi_{02} + \varphi_{12} y_{t-1} + \varphi_{22} y_{t-2} + v_t$

$\varphi_{22} = -0{,}35$

$\rho_1 = \varphi_{12}\rho_0 + \varphi_{22}\rho_1$

$\rho_2 = \varphi_{12}\rho_1 + \varphi_{22}\rho_0$

$$\begin{pmatrix} \rho_0 & \rho_1 \\ \rho_1 & \rho_0 \end{pmatrix}\begin{pmatrix} \varphi_{12} \\ \varphi_{22} \end{pmatrix} = \begin{pmatrix} \rho_1 \\ \rho_2 \end{pmatrix}$$

$$\varphi_{22} = \begin{vmatrix} \rho_0 & \rho_1 \\ \rho_1 & \rho_2 \end{vmatrix} \Big/ \begin{vmatrix} \rho_0 & \rho_1 \\ \rho_1 & \rho_0 \end{vmatrix} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$$

φ_{33}:

$\rho_1 = \varphi_{13}\rho_0 + \varphi_{23}\rho_1 + \varphi_{33}\rho_2$

$\rho_2 = \varphi_{13}\rho_1 + \varphi_{23}\rho_0 + \varphi_{33}\rho_1$

$\rho_3 = \varphi_{13}\rho_2 + \varphi_{23}\rho_1 + \varphi_{33}\rho_0$

$$\begin{vmatrix} \rho_0 & \rho_1 & \rho_1 \\ \rho_1 & \rho_0 & \rho_2 \\ \rho_2 & \rho_1 & \rho_3 \end{vmatrix} = 0$$

$$\varphi_{33} = 0$$

$$\rho_3 = \underset{a_1}{1{,}2}\,\rho_{t-1} \underset{a_2}{-0{,}35}\,\rho_{t-2}$$

$$y_t = \varepsilon_t + 0{,}5\,\varepsilon_{t-1}$$

$$E y_t = 1 \qquad Var\, y_t = 1{,}25\,\sigma_\varepsilon^2 = \gamma$$

$$\gamma_1 = E\left[(\varepsilon_t + 0{,}5\varepsilon_{t-1})(\varepsilon_{t-1} + 0{,}5\varepsilon_{t-2})\right] =$$

$$= 0{,}5\,\sigma_\varepsilon^2 \Rightarrow \rho_1 = \frac{0{,}5}{1{,}25} = \frac{0{,}1}{0{,}25} = \frac{4}{10}$$

$$\gamma_2 = E\left((\varepsilon_t + 0{,}5\varepsilon_{t-1})(\varepsilon_{t-2} + 0{,}5\varepsilon_{t-3})\right) = 0$$

$$\rho_2 = 0$$

$$\varphi_{11} = \rho_1$$

$$\rho_{22} = \begin{vmatrix} \rho_0 & \rho_1 \\ \rho_1 & \rho_2 \end{vmatrix} \Big/ \begin{vmatrix} \rho_0 & \rho_1 \\ \rho_1 & \rho_0 \end{vmatrix} = \frac{-\frac{16}{100}}{1 - \frac{16}{100}} = \frac{-16}{84} = -\frac{4}{21}$$

$$\varphi_{33} = \begin{vmatrix} \rho_0 & \rho_1 & \rho_1 \\ \rho_1 & \rho_0 & \rho_2 \\ \rho_2 & \rho_1 & \rho_3 \end{vmatrix} \Big/ \begin{vmatrix} \rho_0 & \rho_1 & \rho_2 \\ \rho_1 & \rho_0 & \rho_1 \\ \rho_2 & \rho_1 & \rho_0 \end{vmatrix} = \frac{\rho_1^3}{1 - 2\rho_1^2}$$

$$y_t = \varepsilon_t + 0{,}5\varepsilon_{t-1} + 0{,}5\varepsilon_{t-2}$$

$$\gamma_0 = \left(1 + 0{,}25 + 0{,}25\right)\sigma_\varepsilon^2 = 1{,}5\,\sigma_\varepsilon^2$$

$$\gamma_1 = E\left[(\varepsilon_t + 0{,}5\varepsilon_{t-1} + 0{,}5\varepsilon_{t-2})(\varepsilon_{t-1} + 0{,}5\varepsilon_{t-2} + 0{,}5\varepsilon_{t-3})\right] =$$

$$\gamma_2 = E[(\varepsilon_{t-2} + 0.5\varepsilon_{t-3})(\varepsilon_{t-2} + 0.5\varepsilon_{t-3})] =$$

$$= E[\varepsilon_{t-2}^2] + 0.25E[\varepsilon_{t-3}^2] =$$

$$= 0.25\sigma_\varepsilon^2$$